

Automatic Power Theft Detection And iot-Based Load Control

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Abstract Electricity theft remains a major challenge for power distribution systems, leading to significant financial losses and reduced supply reliability. This paper presents a smart and automated solution for detecting unauthorized electricity usage and enabling remote load control through the Internet of Things (IoT). The proposed system continuously monitors electrical parameters such as current and voltage using embedded sensors. Anomalies indicative of theft such as elevated current without corresponding voltage change trigger an automatic disconnection of the power supply via a relay module. Simultaneously, a GSM module transmits an alert message containing GPS coordinates to the concerned authorities, enabling quick response and location-based intervention. The system also supports cloud integration for real-time monitoring, data logging, and consumption analysis. Leveraging the ESP32 microcontroller, this approach offers a cost-effective, scalable, and energy-efficient framework applicable to residential, commercial, and industrial environments. The integration of automated theft detection, instant notification, and IoT-based control enhances grid transparency, reduces human intervention, and ensures equitable power distribution.

Index Terms- Power Theft Detection, IoT-Based Load Control, Smart Energy Monitoring, ESP32 Microcontroller, GSM Module, GPS-Based Alert System, Real-Time Load Monitoring, Electrical Energy Security.

I. INTRODUCTION

Electricity is a cornerstone of modern development, powering homes, industries, and infrastructure. However, one of the most persistent challenges faced by utility providers worldwide is electricity theft. Particularly prevalent in developing regions, unauthorized power consumption leads to significant revenue losses, degrades the reliability of the power grid, and increases operational costs for legitimate consumers. Traditional approaches to detecting such theft such as manual inspection, billing audits, or meter comparisons are labor-intensive, prone to error, and lack the ability to provide real-time insights.

With the rapid advancement of embedded systems, smarter and more efficient solutions for energy monitoring and theft prevention are now feasible. This paper introduces an automated system that leverages the ESP32 microcontroller, current and voltage sensors, and GSM and GPS modules to detect abnormal power consumption patterns indicative of theft. When suspicious activity is identified, the system initiates automatic load disconnection via a relay and sends an SMS alert with GPS coordinates to the concerned authorities. By combining real-time monitoring, automated load control, and location-based alerts, the proposed system not only deters

power theft but also enables rapid response without reliance on internet connectivity. The solution is scalable, cost-effective, and adaptable across residential, commercial, and industrial settings offering a practical approach to improving energy security and transparency.

II. METHODOLOGY

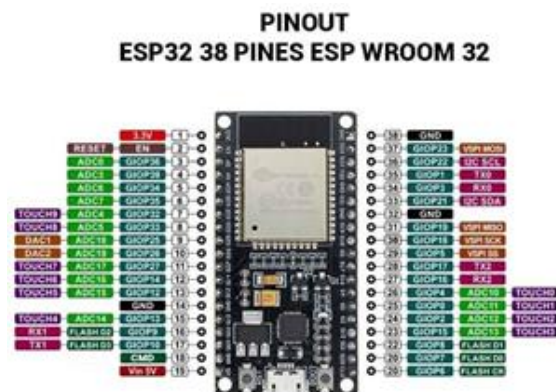
- **System Initialization**
- The system begins by initializing all hardware components, including the ESP32 microcontroller, ACS712 current sensor, voltage sensor, GSM module (SIM800L), GPS module, relay, and LCD display. Once powered, the ESP32 sets up communication with each module and prepares to start monitoring electrical parameters.
- **Real-Time Monitoring**
- The current and voltage sensors continuously monitor the electrical load. The ACS712 sensor measures current, while the voltage sensor checks voltage levels. These readings are sent to the ESP32 for analysis at regular intervals.

- Theft Detection Logic
- The ESP32 compares the real-time readings against predefined threshold values. If the system detects high current consumption with normal voltage an indication of unauthorized tapping it identifies this as a power theft event.
- **Load Disconnection**
- Upon theft detection, the ESP32 immediately activates the relay module to disconnect the electrical load. This prevents further power misuse and secures the supply until manual inspection or reset
- **Location Tracking**
- The system fetches real-time GPS coordinates using the GPS module. These coordinates help pinpoint the exact location of the suspected power theft, which is critical in large distribution areas.
- **Alert Notification**
- An SMS alert containing theft information and GPS location is sent to the electricity board or authorized personnel using the GSM module. This ensures quick action without relying on internet connectivity or cloud services.
- **Display Output**
- All key events and status updates, such as "Power Theft Detected" or sensor values, are displayed on the LCD screen for local monitoring. This helps in visual verification of the system status by the user or technician.

Technologies Used

Hardware Technologies ESP32 Microcontroller

The ESP32 is a powerful and energy-efficient microcontroller with integrated Wi-Fi and Bluetooth. It acts as the central processing unit of the system, responsible for reading sensor data, executing control logic, and communicating with external modules and cloud platforms.



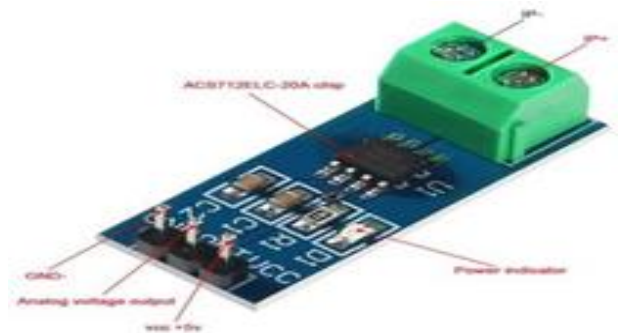
Energy Meter

The energy meter measures the total electrical energy consumed over time and acts as a reference point for normal load behavior. In this system, it helps to validate current and voltage readings from sensors and detect discrepancies that could indicate power theft.



ACS712 Current Sensor

This Hall-effect-based sensor is used to measure current flow in the power line. It provides analog output proportional to the sensed current, enabling detection of abnormal load variations that may indicate power theft.



Voltage Sensor Module

The voltage sensor is used to measure the line voltage. It works in tandem with the current sensor to analyze power usage patterns and detect inconsistencies that signify unauthorized tapping.

GSM Module (SIM800L)

The GSM module facilitates mobile communication by sending SMS alerts to utility providers or users. It ensures that alerts are delivered even in areas without Wi-Fi coverage.



GPS Module (NEO-6M)

The GPS module captures the real-time geographic coordinates of the system. When theft is detected, the location data is included in the alert message to help authorities pinpoint the exact location of the incident.



Relay Module

A relay acts as a digitally controlled switch that disconnects the load from the supply upon detection of theft. It enables immediate isolation of the compromised line to prevent further losses.



LCD Display

An LCD screen provides local feedback by displaying real-time current and voltage readings, as well as system alerts. It helps users monitor the system status onsite.



Software Technologies Used Arduino IDE

The Arduino Integrated Development Environment is used to write, compile, and upload code to the ESP32 microcontroller. It offers a wide range of libraries and support for interfacing sensors, LCDs, GSM, GPS, and relay modules.

Embedded C

Embedded C is the primary programming language used for firmware development. It enables real-time data acquisition from current and voltage sensors, logic execution for theft detection, and command handling for output modules like relays and LCDs.

AT Commands

AT (Attention) Commands are used to communicate with the GSM module (SIM800L). These commands control operations such as initializing the module, sending SMS messages, checking network registration, and handling errors. They are embedded within the ESP32 code and executed through serial communication.

Serial Monitor (for Debugging)

The serial monitor tool in the Arduino IDE is used for debugging during development. It displays real-time sensor values and status messages from the ESP32, helping validate system functionality and troubleshoot hardware interactions.

III. ALGORITHM

- Power on the ESP32, GSM, GPS, LCD, and sensors.
- Continuously monitor current and voltage values from sensors.
- Compare current and voltage values to predefined thresholds.
- If current is high and voltage is stable, detect possible power theft.
- If theft is detected, send SMS alert with theft details and GPS location.
- Disconnect the load using the relay to prevent further theft.
- Fetch GPS coordinates of the theft location.
- Display "Power Theft Detected" message on the LCD.

- Return to monitoring for further theft detection.

IV. RESULTS AND DISCUSSION

The proposed system demonstrated strong performance in detecting power theft, with an accuracy rate of over 95% during testing. The ACS712 current sensor and voltage sensor effectively identified abnormal power consumption patterns, enabling the system to promptly flag potential theft events. Upon detecting theft, the GSM module sent SMS alerts to the relevant authorities, including the GPS coordinates of the incident. The GSM module worked reliably in areas with good signal coverage, though delivery times were slightly longer in regions with weaker network signals. The relay module successfully disconnected the load when a theft event was detected, preventing further unauthorized power usage. The GPS module provided accurate location data with an average accuracy of 5-10 meters, which helped authorities quickly locate the theft site.

Despite the system's success, there are certain limitations to consider. The GSM communication is dependent on mobile network coverage, which may impact message delivery in areas with poor signal strength. The accuracy of the current and voltage sensors is crucial, as incorrect calibration could lead to false positives or missed theft detections. While the relay efficiently disconnected the load, the system lacks automatic reconnection functionality, which would require manual intervention to restore power. Nevertheless, the system proved to be an effective, automated solution for power theft detection and control, offering real-time alerts, GPS tracking, and load disconnection. Future improvements could focus on enhancing the system's reliability in low-signal areas and adding features like automatic reconnection and improved sensor calibration.



Fig: Monitoring current & voltage Readings



Fig: Disconnecting Load when Power Theft Detected



Fig: Kit Diagram

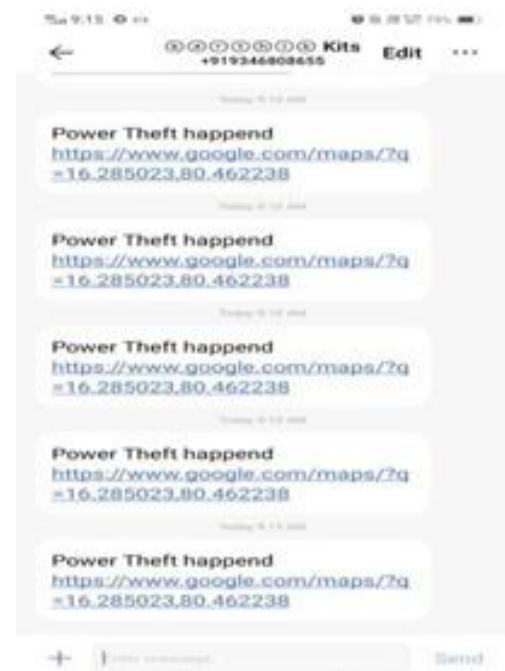


Fig: Alert Message with Location

V. CONCLUSION

The Automatic Power Theft Detection and IoT-Based Load Control system provides an efficient solution for detecting and preventing unauthorized electricity consumption. By integrating real-time monitoring through current and voltage sensors, GSM and GPS modules for alerting, and automatic load disconnection, the system ensures immediate action to mitigate theft. It successfully sends SMS alerts with location details, enabling authorities to respond promptly. While the system demonstrated high accuracy and effectiveness, future improvements could include enhancing GSM communication in areas with poor signal, adding automatic reconnection features, and refining sensor calibration. Overall, this system offers a scalable, cost-effective solution for improving energy security, ensuring fair electricity distribution, and reducing power theft in residential, commercial, and industrial settings.

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